Tech News

Technology Updates

Autonomous vehicles could benefit health if cars are electric, shared

The Barcelona Institute for Global Health (ISGlobal) has taken part in a study that analysed potential risks and benefits of autonomous vehicles (AVs) for public health. Conclusions of the study indicate that this new type of mobility could benefit public health if cars are electric and the model used is based on ride-sharing.

The research team found that while this mode of transportation could increase some health risks, including air pollution, noise and sedentary behaviour, if properly regulated, it would likely reduce morbidity and mortality from motor vehicle crashes and may promote healthier urban environments.

Representational image

David Rojas, first author of the paper and a researcher at ISGlobal and Colorado State University, explains the current situation, "At the international level, we are still seeing very little research or planning by authorities in anticipation of the advent of these new transport technologies, despite the fact that AVs have the potential to significantly modify our cities and the way we travel. And this innovative autonomous technology will also have an impact on public health.

"The advent of AVs may result in either health benefits or risks, depending on a number of factors, such as how the technology is implemented, what fuel and engines are used, how self-driving cars are used and how they are integrated with other modes of transport."

Smart intersections could reduce autonomous car congestion

In the not-so-distant future, city streets could be flooded with autonomous vehicles (AVs). Such self-driving cars can move faster and travel closer together, allowing more of them to fit on the road, potentially leading to congestion and gridlock on city streets.

A new study by Cornell researchers developed a first-of-its-kind model to control traffic and intersections to increase car capacity on urban streets, reduce congestion and minimise accidents.

Oliver Gao, professor of civil and environmental engineering and senior author, says, "If you have all these autonomous cars on the road, you will see that roads and intersections could become the limiting factor."

The researchers' model allows groups of autonomous cars, known as platoons, to pass through one-way intersections without waiting, and results of a mirco-simulation showed it increased the capacity of vehicles on city streets up to 138 per cent over conventional traffic signal systems, according to the study. The model assumes only autonomous cars are on the road; Gao's team is addressing situations with a combination of autonomous and human-driven cars in future research.

Gao adds, "Instead of having a fixed green or red light at the intersection, these cycles can be adjusted dynamically. And this control can be adjusted to allow for platoons of cars to pass."